

DAVIS WHALEY Atlanta, Georgia | 859-229-0155 | daviswhaley@gmail.com LinkedIn: [linkedin.com/in/davis-whaley-3175a9212](https://www.linkedin.com/in/davis-whaley-3175a9212)

SUMMARY Data Scientist and USCF National Chess Master with a Double Master's in Psychology. Expertise in **Trustworthy AI**, unsupervised deep learning, and auditing human-in-the-loop taxonomies. Engineered research-grade manifold audits for high-stakes scientific domains (LIGO Astrophysics & Histopathology), uncovering systematic labeling errors using Computer Vision and Density Clustering. Proven track record of managing large-scale datasets and executing end-to-end research cycles from raw data to deployed cloud dashboards.

SELECTED PROJECTS

The Pathologist Uncertainty Auditor (BreaKHis) | [[GitHub Link](#)] [[Live App Link](#)]

- **Objective:** Engineered a "Trustworthy AI" pipeline to audit diagnostic uncertainty in 7,909 breast cancer histology images, challenging the discrete binary nature of medical taxonomies.
- **Methodology:** Utilized **ResNet-50** for feature extraction and **UMAP/HDBSCAN** to map the "Ambiguous Manifold," identifying 47.8% of the dataset as "morphological noise" rather than discrete categories.

LIGO Gravity Spy Glitch Explorer | [[Gravity Spy Github](#)] [[Interactive Dashboard](#)]

- **Engineered a deep-learning audit pipeline** (PyTorch, ResNet-18) to extract 512-D morphological features from 34,332 gravity-wave glitch spectrograms.
- Identified a **28.5% human-labeling error rate** by mapping 1-D morphological manifolds (UMAP/HDBSCAN), revealing systematic failures in discrete classification taxonomies.
- Built a public-facing interactive dashboard using **Plotly** and **GitHub Pages** to visualize high-dimensional data clusters with real-time image previews.

Gravitational-Wave Transient Catalog (GWTC) | [[GWTC Github](#)]

- **Built an interactive data visualization dashboard in Power BI** using publicly released LIGO–Virgo gravitational-wave catalogs (GWTC-1 through GWTC-4), exploring black hole mergers and signal attributes.

TECHNICAL SKILLS

- **Languages & Tools:** Python (Pandas, PyTorch, Plotly, NumPy, Scikit-learn, Streamlit), SQL, R, Power BI, Excel, SPSS, Git/GitHub.

- **Modeling & Analysis:** Computer Vision (ResNet/CNNs), Manifold Learning (UMAP), Clustering (HDBSCAN), Transfer Learning, Time Series Analysis (ARIMA, LSTM, VAR), Predictive Modeling.
- **Research Methods:** Experimental Design, Mixed-Methods (Qual/Quant), "Think Aloud" Protocols, Data-Centric AI Auditing.

PROFESSIONAL EXPERIENCE

Freelance Data Analyst & Programmer | Remote 2022 – Present

- **Key Project: Behavioral Time-Series Forecasting**
 - Engineered a predictive modeling pipeline using **Python (Keras/TensorFlow)** to analyze longitudinal behavioral metrics.
 - Used **Vector Autoregression (VAR)** and **LSTM** neural networks to forecast trends, performing extensive feature engineering on raw time-series data.

Graduate Research Fellow (PhD Track) | Georgia State University | Atlanta, GA 07/2019 – 12/2021

- **Qualitative Leadership:** Ran 60+ "Think Aloud" interviews to extract nuanced insights into reading comprehension. Analyzed behavioral data using SPSS for statistical modeling and inference generation.

Graduate Research Assistant | University of Memphis | Memphis, TN 08/2017 – 05/2019

- Managed a large-scale experimental study (N=800) assessing reading comprehension under varying conditions.
- Built data cleaning pipelines in **R** to handle raw post-test datasets, identifying key variables influencing user performance, validating hypotheses regarding science-text comprehension. Presented findings at 2019 Psychonomics Conference.

EDUCATION

- **M.S. Educational Psychology** | Georgia State University (2021)
- **M.S. General Psychology** | University of Memphis (2019)
- **B.A. Spanish** | University of Kentucky (2010)

AWARDS & RECOGNITION

- **USCF National Chess Master | 11-Time Kentucky State Chess Champion**

